JY-CRPG-BENCH: A LONG-HORIZON BENCHMARK FOR VISION-LANGUAGE MODELS IN A CHINESE OPEN-WORLD RPG

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ABSTRACT

A long-horizon task requires a model to perceive its environment, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language model inside 金庸群俠傳, the 1996 commercial open-world role-playing game *Heroes of Jin Yong* running under DOS emulation, and scores it against the ending the game defines, which requires recovering the fourteen books of the Jin Yong canon and takes a first-time human player tens of hours. The environment is hard for current vision-language models on several axes at once, with a 320×200 framebuffer of 1996 pixel art, a $45°$ isometric projection in which no key moves straight up the screen, Traditional Chinese rendered as 16-pixel bitmap glyphs, and objectives stated only inside its fiction. Every model plays through the same brief and control surface, with the frame as its observation and keys as its actions, and with no memory scaffold and no access to game state, so a run measures the model. Progress is read from the memory of the emulated machine and from the save the game writes for itself, so no milestone can be earned by a change on the screen alone, and one instrument spans both phases of play, a ladder of state-verified milestones while models learn to move and completion time once they finish. We report 12 frontier models against a seeded random baseline under a 60-minute budget: 10 reach the world map, one holds the compass, and none gains experience. We release JY-CRPG-BENCH as a live, replayable public leaderboard at https://github.com/hanxiao/jy-crpg-bench.

1 INTRODUCTION

Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). A long horizon is a property of the task, the number of dependent steps its completion requires (Chen et al., 2026), and its evaluation takes two forms. Episodic benchmarks set a goal and score its completion; continuing benchmarks give the model an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). An open-world game offers both, an ending that takes tens of hours to reach and a record of everything that accumulates before it, and adds a requirement the business simulations do not: the model has to recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments with a language-only mode (Paglieri et al., 2025), TextQuests removes perception entirely (Phan et al., 2025), ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026), VideoGameBench detects progress by matching the screen against walkthrough checkpoints (Zhang et al., 2025a), and MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed model is asked to do.

(a) the scene every session starts in
(b) the overworld the scenes hang off
(c) the four axes the game moves along
(d) dialogue, 16-pixel glyphs
(e) the menu inside a scene
(f) the record the benchmark reads
(g) the bag, one item described at a time
(h) the outdoor menu and the party
(i) a turn of battle on the grid

Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four axes the movement keys run along (c), the three kinds of text panel through which every objective, choice and statistic is stated (d–f), the bag (g), the outdoor menu with the party list it opens (h), and a turn of battle with the menu and record of the acting character (i).

JY-CRPG-BENCH runs a single deep environment, following the precedent of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳, a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and reputation, and an ending that requires recovering fourteen books scattered across the world. A first human playthrough takes tens of hours. The model receives the framebuffer and a key vocabulary through one fixed, minimal harness, and nothing else, so what varies between runs is the model.

We measure against that ending. The fourteen books sit behind items, factions and martial arts that have to be acquired in order, in a world that reports nothing about itself, and the game records whether they have been recovered. A model that finished would have held one plan across tens of hours of self-directed play, which is the capability long-horizon evaluation exists to detect. The instrument therefore spans both phases of play: a ladder of state-verified milestones orders the field while models are learning to move through the world, and completion time orders it once they finish, so the benchmark gains a speedrun ranking where a pass-or-fail score would saturate.

Three properties make the environment hard for current models, and Figure 1 shows the screens they apply to. First, perception is difficult on several documented axes at once. Downsampling costs vision-language models accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), out-of-distribution images degrade recognition even of familiar categories (Lin et al., 2026), text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024); a 320×200 frame of 1996 pixel art carrying 16-pixel Traditional Chinese glyphs sits at the hard end of all five. Second, the world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals, and a model that reasons in

Table 1: Interactive agent benchmarks. *Observation* is what the model receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save file

screen coordinates walks in circles. Third, progress is self-directed. The game keeps no quest log, draws no objective marker and reports no score. The brief supplies the opening route, from the first room to the compass, and a manual of the controls; from the compass on, deciding where to go and what is worth doing is part of the problem, the goal-setting requirement that ARC-AGI-3 places at the centre of agentic intelligence (ARC Prize Foundation, 2026).

Measuring such an environment from the screen is unreliable, and recent work says so: GameWorld names heuristic verification as the obstacle to standardised game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position inside the emulator. The benchmark reads progress from the memory of the running machine, where the game maintains its character array, and from the save file the game writes when asked, which holds the party roster and the world position. The model has no call that returns any of this, and nothing read from the screen enters a score.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on a commercial open-world role-playing game whose difficulty comes from named and independently checkable properties of the environment: low-resolution perception of 1996 pixel art, isometric control, Traditional Chinese grounding, a keyboard action space of which the game answers a small subset, and persistent character state that only coherent play advances.
2. We score progress in the terms the game keeps. Character records, the party and the fourteen books are read from emulator memory and from a save the game writes for itself, validated against the archives the release ships, and measurement never writes to the machine it observes. Every rung of the ladder is an event the model has to cause, and the terminal rung is the ending the game defines.
3. We evaluate 12 frontier models under the same harness and against a seeded random baseline under a wall-clock budget, and release the benchmark, its harness and every recorded run as a public, replayable leaderboard.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. The next section reviews game environments as agent benchmarks and the measurement problem that long horizons create, Section 3 details the environment, the interface, the session protocol and the measurement, and Section 4 reports the field: how much of the ladder is open, how far it separates models, how the budget is spent, and what the replays show.

2 RELATED WORK

2.1 GAME ENVIRONMENTS AS AGENT BENCHMARKS

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is the nearest benchmark to ours, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025a). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. The nearest precedents in horizon are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025). Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025). The PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). PokeGym builds a long-horizon benchmark on the open-world game Pokémon Legends: Z-A with no access to game state (Zhang et al., 2026a), and StarBench one on the turn-based Honkai: Star Rail (Zhang et al., 2025b). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the deepest environment in use for learned agents (Küttler et al., 2020). JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory and from the save file of the game, and in scoring against the completion condition the game defines.

2.2 Long-Horizon Evaluation and Its Measurement

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025), ARC-AGI-3 isolates adaptation to novel tasks by excluding language and external knowledge (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025).

Measurement is where long horizons strain evaluation. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), an audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder takes the same approach, read from machine state so that dense signals cannot be mistaken for progress.

3 JY-CRPG-BENCH

3.1 Environment

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Every scene carries an entry condition, and at the start all of

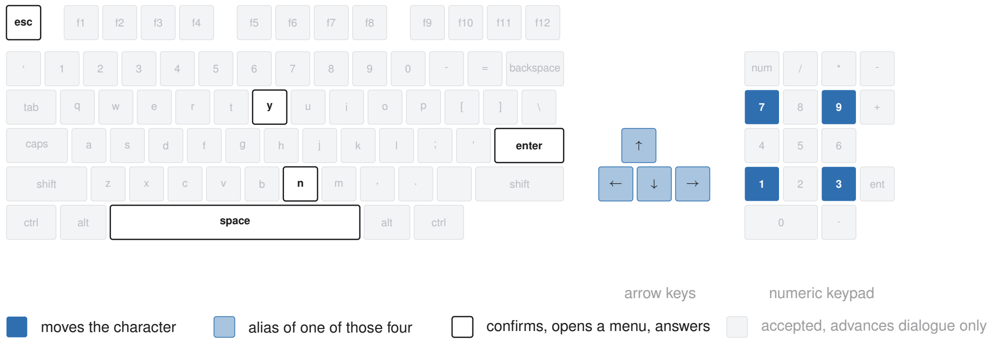

Figure 2: The keyboard the API exposes. The nine keys the game answers outside dialogue are picked out, the arrow keys are aliases of the four movement keys, and any key advances a dialogue.

them are closed except the home, five inns and the house of the hermit 南賢; events open the rest, and the opening sets their order.

The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit, whose long conversation opens the house of the first companion, and a cabinet beside him holds the compass, the only source of coordinates the game offers. The companion joins after a yes-or-no prompt at which the confirm key means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force, and the first scripted fight opens by draining it. An automatic mode hands the character to the AI of the game, which alternates striking and resting and loses to an opponent who heals. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one, and a lost fight ends the game, which then offers nothing but its own load menu.

Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, and the character sheet on screen displays the same record the benchmark reads from memory. We selected this title for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely neglected (Tam et al., 2025), so cross-lingual grounding is a central obstacle. It is menu-driven, so a model that deliberates between actions is not structurally disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025a). Its progression state is rich enough to measure at the level of individual character statistics. Its $45°$ isometric projection makes the mapping from screen to action something the model must infer.

3.2 Observation and Action Space

The model observes the raw framebuffer at 320×200 and acts by pressing keys. A scored session serves the native frame without upscaling, annotation, set-of-marks overlay or accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, with 130 key names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads its movement from the scrolling of the background loses count of the tiles it walked there. Canopies and roofs are drawn over the character, and the entrance of a building is a single tile on one face of its wall, so reaching a door means walking around the compound to the side the door is on.

The one action presses a key, or a list of keys in order, holding each for a requested number of emulated frames, ten by default, and returns once the screen has reacted and held still, waiting through a scene transition if one starts, so the frame a model receives is the one to act on. There is no call that advances the game without input, since the game only moves on input, and the hold is counted in frames because the game samples its keyboard once per game-loop iteration. A scored session answers the first four calls of Table 2: the screen, the brief, the key names and the action. Three further calls save, list and restore emulator snapshots for development, and a scored session

Table 2: The seven calls of the control surface. A scored session answers the first four; the three snapshot calls serve development and are refused in a scored session.

Call	Body	Meaning
`GET /api/screen`	–	the current frame
`GET /api/help`	–	the brief, in English or Chinese
`GET /api/keys`	–	every accepted key name
`POST /api/key`	`{"key":"kp3","hold":10}`	press one key, or a list of keys in order
`GET /api/slots`	–	emulator snapshots on disk
`POST /api/save`	`{"name":"..."}`	take a snapshot
`POST /api/load`	`{"name":"..."}`	restore a snapshot

refuses them, so a run cannot be rewound from outside the game. The save and load menu of the game stays open to the model as to a human player, so a model that loses a fight can reload the last save the game wrote, and the recording shows it doing so.

3.3 Harness and Session Protocol

The benchmark supplies a minimal harness. The brief is the whole instruction the model receives: the calls that constitute play, the rules of a run, the opening route as far as the compass, and a manual of the controls and systems of the game, served in English and Chinese with the on-screen Traditional Chinese terms kept verbatim, so the language a model reasons in is not itself a handicap. An action returns only its metadata, so the model looks at the screen when it decides to, and there is no memory scaffold, no map, no pathfinder and no access to the state of the game. This is what makes the leaderboard a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026b).

Table 2 lists the control surface in full, and the same brief is published at `agents.md` for a client that creates its own sessions. The reference harness reads the Chinese brief by default, the record of a run keeps the language of every brief it fetched, and the manual states that the movement keys run along the isometric diagonals. A scored session is refused every field that would report its own progress until the run ends, so a model cannot query its own score.

A session is created under a declared name with a playtime, 60 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix B, with its save slots reset to a new game, so no run inherits the progress of another. A session ends only when its budget lapses or the game is completed, and there is no inactivity cutoff, so a long pause between keys costs the model nothing but budget. When the budget lapses the session renders its recording to a time-compressed video, computes its metrics, and appends itself to the public catalogue with the video and its keypress timeline, so the behaviour behind a score can be reviewed. Figure 3 shows the apparatus, and Appendix A lists the session parameters.

3.4 State-Verified Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the opening machine image and the archives the release ships, and we locate the same array in the live emulated machine by anchoring on one character name and confirming the two neighbouring records at the expected stride, since the name recurs in the image and the layout moves between runs. The bag in front of that array is the working copy and changes the moment an item is taken, so the item count, the compass and the books are read from it after every action and again from every save the game writes. A rung, once seen, is kept, so an item picked up and later spent still credits it. Reading is passive: the benchmark serialises the machine and never loads state into it. The only keys it presses on its own are those of the save described next, which take the game a few seconds of menu time that is not charged to the run and leave the screen as they found it.

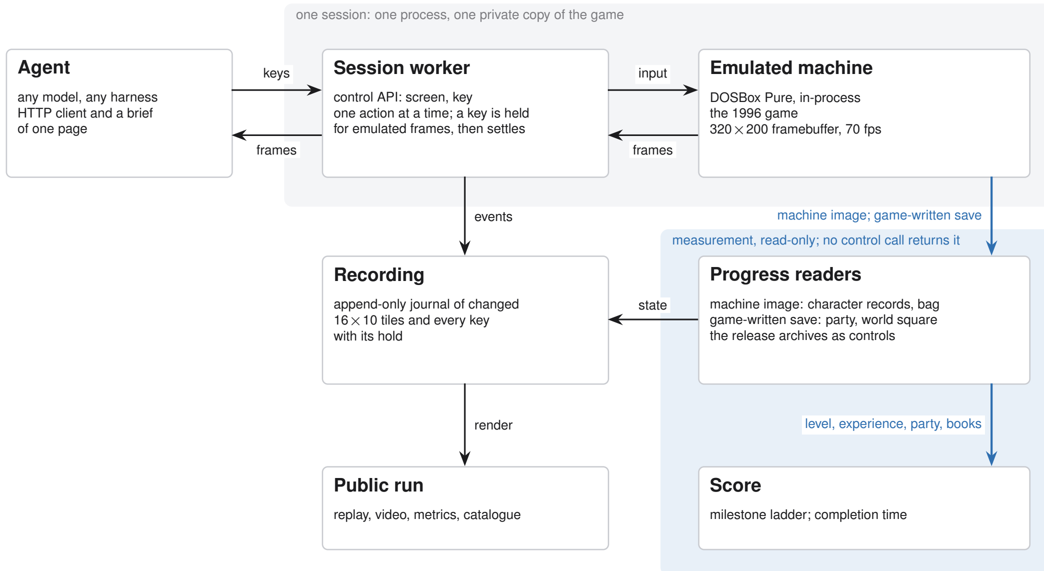

Figure 3: One process per session loads the game core and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine and from the save the game writes, on a path no control call reaches.

For the party roster and the world position the benchmark reads the save the game writes. The game offers its save menu on the world map alone and announces this through the height of the menu, which stands at six rows on the world map and four in a scene, so an attempt opens the menu, reads its height, and saves only where six rows appear. An attempt recurs every two minutes and once more before the budget lapses, and waits for a two-second gap in the keys the model sends. The session is held while the attempt runs, so the model never sees the screen mid-save, and the time of the first save that lands is the time of the crossing. Neither the archive nor anything read from it is reachable through the control surface, so a save that lands is itself the evidence that the party has crossed onto the world map. The session record also keeps a screen fingerprint of the world map for the replay, and it never enters a score.

Progress is summarised as eight milestones read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021), and Table 3 lists them with the evidence each rests on. Only the first concerns the harness; the rest are the numbers the game keeps about itself. The rungs fall into two phases that the game itself separates. The opening needs no fight: the chest in the first room, the exit tile, the compass in the cabinet of the hermit, and the companion whose house that conversation opens and who joins through dialogue are reached by walking, reading and answering. The campaign begins with the first fight, the first source of experience and levels, and runs through the chains of items, factions and fights that the books sit behind. The default budget is sized to the opening and the extended budgets to the campaign. The rungs are listed in the order the opening imposes, but a run is credited with every rung it reaches, so the count does not depend on the path taken.

The ladder runs to the ending of the game. The fourteen books, named in Appendix C, are what the game requires before its final sequence can be played, and the archive records every book held, so the count of books extends the ladder beyond its last rung, and the first read that finds all fourteen held records the completion time. A session that completes carries that time, which orders a field in which every entrant passes, so one instrument covers both phases and the benchmark keeps measuring after the first model finishes.

The ladder is the fixed score of the benchmark. For the multi-hour budgets the protocol allows, the release adds a second, additive report over the same ground truth: a finer trajectory of the scenes a run reached, the battles it survived, the skills it learned and the books it gathered, out to the ending, with the hard prerequisites of the walkthrough as dependency gates between them. Each component is scored against a fixed denominator with its coverage reported beside it, and the report never

Table 3: The milestone ladder. *Evidence* is the game state that credits the rung and *source* is where the benchmark reads it: the session record of the harness, the bag and character array in emulator memory, or the save the game writes on the world map.

Rung	Evidence	Source
The opening, reached without a fight		
acted	a key event was sent	session record
picked something up	any item count above its opening count	bag in memory
reached the world map	the game wrote its save	save
holds the compass	the compass in the bag	bag in memory
recruited a companion	a party roster longer than one	save
The campaign, from the first fight on		
gained experience	experience above zero	character record
reached level 2	level above one	character record
holds one of the fourteen	a book in the bag or carried by the protagonist	bag and record

rescores the ladder, so a long run holding no book still leaves an ordered record of how far its plan carried. Its checkpoint contract and component weights are released with the benchmark.

3.5 Effort

Beside the ladder the session records effort. The wall clock of every decision call is kept on the server, so the played time and the think time between calls are measured for any client, and the reference harness files the tokens a run spent, with the model and its thinking level, when the run ends. Effort is shown beside a run and never ranks it.

A seeded random policy plays the same budget through the same control surface, drawing from 13 keys the brief teaches, weighted towards movement and confirmation, and never reading the screen, following the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). The rungs it reaches mark the progress available without reading the screen and are the floor of the ladder. Because it is seeded, its whole key sequence is recomputed offline and checked against the journal of keys the machine received.

4 Evaluation

The field at submission is 12 models from 5 vendors at the 60-minute budget, and a seeded random baseline. A run is listed under the name declared when its session was created, and a model is reported by one session: 2 models were run more than once, and each is reported by the longest session it played. The 2 shorter sessions stay in the snapshot, and 1 of them never sent a key. No session filed a token report, so effort is reported as wall clock, and the 2 sessions that fetched the brief from the session did so in English. Every session is public on the leaderboard with its video, replay timeline and metric record, and the catalogue snapshot the paper is generated from is committed beside the paper source. Appendix D lists the runs by name. The subsections that follow report how much of the ladder is open, how far it separates models, how the budget is spent, and what the replays show.

[The present field reports one run per model. A sweep of repeated runs per model under the fixed harness, and a human reference run under the same budget and interface reported to the human-baseline checklist (Wei et al., 2025), are the two additions the field most needs.]

4.1 Headroom

Every rung is measured, and the highest rung any session reaches at this budget is the compass. 8 of 12 sessions pick something up, 10 reach the world map, and 2 spend the whole budget in the compound they started in. One session, `gpt-6-astra`, holds the compass and is the only session to clear a rung beyond the world map. No session recruits a companion, gains experience or holds a book: all 12 character records remain at level 1 with 0 experience and 1 skill, as they were when the game began, so every rung from the companion on is open. Figure 4 shows the ladder by model.

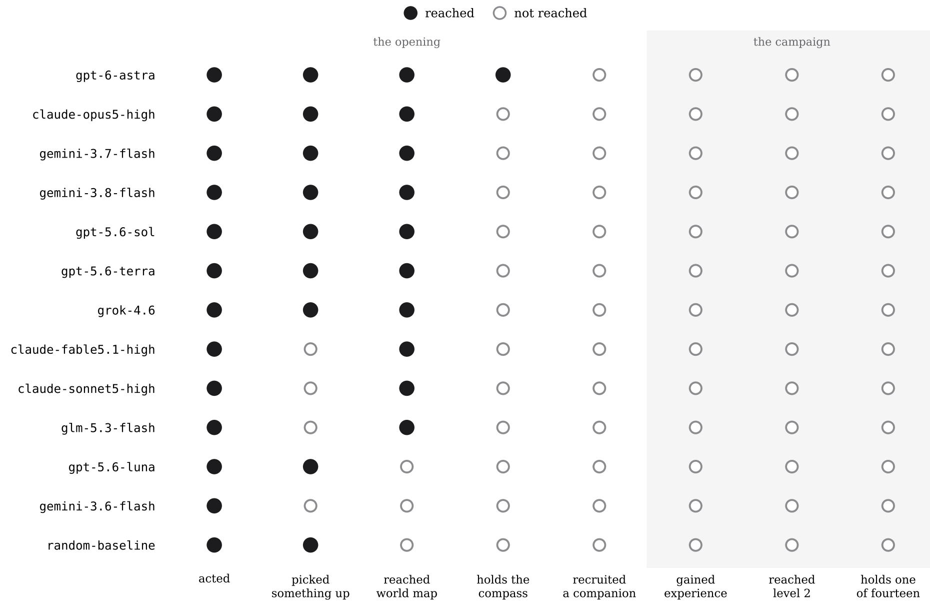

Figure 4: The milestone ladder, one row per model, with filled markers for rungs reached and hollow markers for rungs not reached.

Comparable benchmarks report their fields at a similar point: BALROG places its best model at 1.5% progression on NetHack (Paglieri et al., 2025), and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025a). The distance from the compass to the fourteen books is the headroom the benchmark holds.

4.2 Discrimination

Three rungs divide the field, and they divide it differently. The chest and the exit are both in the first room, and the sessions that searched one are not all the sessions that found the other: 3 crossed without the chest, 1 searched it without crossing, and 7 did both, so the count of rungs summarises what a run achieved and presumes no single path. The random policy reaches 2 of the eight rungs in 60 minutes: it acts, and in a small room a random walk searches the chest within the hour. Its recomputed key sequence matches the journal of the machine key for key, and no save lands during its hour, so the world map is the first rung a random walk does not reach, and the chest rung separates models from one another but not from the floor. The compass divides the field a third time, separating one session from the rest, and above it the field is undivided, since no session recruits the companion. The world-map rung also shows why progress is read from the machine: the save and the screen fingerprint kept for the replay agree on 9 crossings, the save credits 1 crossing the fingerprint missed, and the fingerprint never appears without a save behind it.

4.3 Budget

Every session ran to its budget. The crossing onto the world map is timed by the game: the first save it writes there trails the crossing by at most the two-minute cadence of the attempts, and by that clock the 10 crossings fall between 9 and 41 minutes into the session, with a median of 26. The session that went on to the compass crossed at 9 minutes, after 33 actions, and spent the rest of the hour beyond the exit, so the hour covers the first room and, for one session, the walk south to the cabinet. The budget is wall clock, so time spent deliberating is time not spent playing, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025).

[Long runs that exercise the additive long-horizon report are the measurement that separates a slow model from a stuck one; the protocol and the report already support them, and the field does not yet contain them.]

Figure 5: Frames from the published replays, each with the strip of its video naming the model, the action count and the clock: (a) `gpt-6-astra` reads its coordinates from the compass on the item screen; (b) the same session in the conversation of the hermit; (c) `gemini-3.8-flash` is told where the compass lies and (d) stands at the cabinet as the budget lapses; (e) `glm-5.3-flash` in the yard behind the fence; (f) `claude-fable5.1-high` passing the gate of the hermit; (g) `grok-4.6` inside an inn and (i) `gpt-5.6-sol` in a shop, each talking to the keeper; (h) `gpt-5.6-terra` in the middle of a long key list.

4.4 Failure Modes

The replays show where the hour goes, and Figure 5 collects one frame from each pattern. The session that holds the compass, `gpt-6-astra`, left the compound after 33 actions, walked the path south, and pressed through the conversation of the hermit in 7 minutes at 10 seconds a line before searching the cabinet. It then opened the item screen 9 times to read its coordinates from the compass, the one instrument the brief recommends, and its 42 arrow-key presses all fell inside menus and prompts, never on the map. It entered 11 scenes in the hour, and only 4 sessions entered three or more. `gemini-3.8-flash` reached the same conversation and pressed through its 40 lines at 23 seconds each, so the conversation took 15 minutes and ended at minute 54. The hermit had named the cabinet, 6 minutes of budget remained, and the session ended without the compass.

The compound is left through a yard closed by a fence with one gate, and the yard held several sessions. `glm-5.3-flash` needed 1595 actions to cross, against a median of 130, and reached the world map at minute 37 after long stretches pressed into the fence. It is also the one session that never pressed enter or space, so the chest in the first room, which opens on either, stayed shut. `claude-fable5.1-high` crossed at minute 13 and then spent 1261 actions and 44 minutes on the world map without entering a scene, passing the gate of the hermit repeatedly, since the entrance of a walled compound is one tile on one face and every other face is solid. Two sessions that found an inn and a shop talked to the keeper repeatedly; the reply repeats, and nothing in it advances the opening.

`gpt-5.6-terra` sent lists of up to 100 keys in one action, 22 on average, with 32 actions of fifty keys or more, so it looked at the screen once for every 22 keys it pressed, and it reached the world map credited by the save alone.

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs a 1996 open-world role-playing game, gives every model the same brief and control surface, and scores it in the terms the game keeps about itself. At the 60-minute budget the field of 12 models stays inside the opening of the game: 10 of 12 sessions reach the world map, one holds the compass, and none gains experience, so every rung from the companion to the fourteenth book stands open, and the 2 rungs the random baseline reaches mark the floor a model has to clear. The environment is hard in ways that are named and checkable: the frame is a 320×200 isometric render of 1996 pixel art, objectives are stated only in Traditional Chinese, and the world supplies no score. Progress is read from the memory of the machine and from a save the game writes for itself, so no change on the screen alone can score, and every rung of the ladder is an event the model has to cause. The ladder runs to the ending of the game: the archive records the fourteen books held, so a model that finishes leaves proof of it, and from that point the leaderboard ranks by completion time, with the video and every key of the finishing run on record.

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A SESSION AND INFRASTRUCTURE PARAMETERS

The default budget is 3600 s and is configurable to 24 h. A host of 8 vCPUs and 16 GiB runs up to 24 concurrent sessions, each holding a private copy of the game of about 123 MB. Emulation is DOSBox Pure through libretro at a fixed 26800 cycles, the speed of a 486DX2-66 and period-correct for the game, with no sound card, so the observation is visual only. The published video runs at 8 times speed or faster for long runs, with a strip above the frame naming the actor, the action number, the keys held and the clock, and a timeline file beside it carries every keypress and its hold duration for the replay.

B RELEASE AND LICENSING

The harness code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, loaded at runtime through the libretro C API and redistributed unmodified; the bundled macOS core is built from `schellingb/dosbox-pure` at `7f6e8fb`, and the hosted service ships a Linux build pinned by image digest. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries only the metrics, videos and replay timelines we measured, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

C THE FOURTEEN BOOKS

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or in the four carried slots of the protagonist. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).

D THE FIELD AT SUBMISSION TIME

Table 4 lists the field, one row per model. A model run more than once is listed by the longest session it played, and the catalogue snapshot keeps every session. The live leaderboard at `https://hanxiao.io/jy-crpg-bench/` shows the field and the runs published since.

Table 4: The field, one row per model. *played* is minutes of play, *ended* is how the session ended, *rungs* is the count of rungs reached, and the eight columns mark each rung, ✓ for reached and ○ for not reached. Rows are ordered by rungs reached and then by played time, with the random floor last.

Model	played	ended	rungs	acted	item	map	compass	party	exp	lv 2	book
`gpt-6-astra`	60	time	4/8	✓	✓	✓	✓	○	○	○	○
`claude-opus5-high`	60	time	3/8	✓	✓	✓	○	○	○	○	○
`gemini-3.8-flash`	60	time	3/8	✓	✓	✓	○	○	○	○	○
`gpt-5.6-sol`	60	time	3/8	✓	✓	✓	○	○	○	○	○
`grok-4.6`	60	time	3/8	✓	✓	✓	○	○	○	○	○
`gemini-3.7-flash`	59	time	3/8	✓	✓	✓	○	○	○	○	○
`gpt-5.6-terra`	46	time	3/8	✓	✓	✓	○	○	○	○	○
`glm-5.3-flash`	60	time	2/8	✓	○	✓	○	○	○	○	○
`gpt-5.6-luna`	60	time	2/8	✓	✓	○	○	○	○	○	○
`claude-sonnet5-high`	60	time	2/8	✓	○	✓	○	○	○	○	○
`claude-fable5.1-high`	57	time	2/8	✓	○	✓	○	○	○	○	○
`gemini-3.6-flash`	60	time	1/8	✓	○	○	○	○	○	○	○
`random-baseline`	60	time	2/8	✓	✓	○	○	○	○	○	○